

IT Management Proposal: Enterprise Data Infrastructure Modernisation

Executive Proposal: Enterprise Data Infrastructure Modernisation

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Executive Proposal: Enterprise Data Infrastructure Modernisation

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Author: Lead Systems Architect

Target Audience: IT Management & Steering Committee

Infrastructure Scope: bda-ai-infra

1. Executive Summary

This proposal outlines the strategic modernisation of our Big Data Analytics & AI Infrastructure (bda-ai-infra). We are transitioning from a legacy, static visualisation-dependent architecture (Tableau) to an **API-First** and **Model Context Protocol (MCP)-Ready** ecosystem powered by **Podman** containers.

By replacing proprietary dashboard tools with an open, containerised integration layer, we eliminate recurring licensing costs while transforming our data warehouse into a **bidirectional, intelligent data hub**. Any enterprise software, open-source application, or

AI agent can securely ingest, transform, query, and share data with strict **Fine-Grained Access Control (FGAC)**.

Human-generated data remains the sole authoritative **Single Source of Truth (SSoT)**. All operational data modified or enriched by AI agents is explicitly tagged and segregated within our big data repository using cryptographic provenance contracts (bda_provenance).

2. Core Strategic Pillars: API-Ready & MCP-Ready

2.1 Dual-Render Infrastructure Architecture Blueprint

1. Standalone Production-Ready SVG Vector Graphic (.svg)

BIDIRECTIONAL INGESTION & HUMAN QUARANTINE LAYER External Systems
/ IoT / Partner APIs / User Feeds / Laravel Human Verification Gate

CENTRAL DATA & VECTOR ENGINE (TIER 0 GOLDEN SSoT) Percona Patroni
PostgreSQL 18 + pgvector (NiFi 2.0 Single Authoritative Writer) Metadata Tagging:
Human Verified Golden SSoT vs AI-Enriched RAG Provenance (bda_provenance)

OPEN REST / gRPC APIs Programmatic Data Sharing Gateway Fine-Grained Access
Control (Row & Column Policies) Fusio API Gateway Ingress (Port 8080/443)

MCP SERVER (MODEL CONTEXT PROTOCOL) Native AI Tooling & Context
Gateway Fine-Grained Access Control (Tool Execution & Scope Limits) Python/Node.js
MCP Gateway (Port 8443 / Stdio / SSE)

ENTERPRISE & OPEN-SOURCE CONSUMERS Apache Superset, Metabase, Web
Applications, ERP Systems

LLMS & AUTONOMOUS AI AGENTIC SYSTEMS Context-Aware Inquiries,
Retrieval-Augmented Generation (RAG)

2. Git-Native Mermaid Topology (.mmd)

```
graph TD
    subgraph Ingestion ["Bidirectional Ingestion & Quarantine Layer"]
        Ext["External Systems / IoT / User Feeds"]
        Laravel["Laravel Web Portal (Human Verification Gate)"]
    end

    subgraph Core ["Central Data & Vector Engine"]
        NiFi["Apache NiFi 2.0 (Single Authoritative DB Writer)"]
        Postgres["Percona Patroni PostgreSQL 18 + pgvector\n(Tier 0 Golden SSoT & AI Provenance Tagging)"]
    end

    subgraph Integration ["Integration & API Layer (Fine-Grained Access Control)"]
        REST["Open REST / gRPC APIs\n(Fusio API Server)"]
        MCP["MCP Server\n(Model Context Protocol Tools)"]
    end

    subgraph Consumers ["Consumers & Autonomous Agents"]
        BI["Open BI (Apache Superset)"]
        AI["LLMs & AI Agentic Systems"]
    end

    Ext -->|Ingress Payload| Laravel
    Laravel -->|Verified Human Sign-off| NiFi
    NiFi -->|Cryptographic Persistence| Postgres
    Postgres -->|FGAC SQL Views| REST
    Postgres -->|FGAC Vector & Tool Context| MCP
    REST -->|Programmatic Inquiries| BI
    MCP -->|Native Tool Execution| AI
```

3. Summary Interface & Routing Table

Source Component	Target Component	Port / Protocol / API Ingress	Security Boundary / Access Key	Operational Significance / Flow Description
External Systems / Users	Laravel Verification Gate	HTTPS (443) / REST	Keycloak OAuth2 / Session Auth	Ingests non-IT user feeds and telemetry into quarantine staging directory.
Laravel Verification Gate	Apache NiFi 2.0 Ingest Gate	Staging Spool / Internal Event	Human Signature / Audit Log	Triggers approval workflow upon explicit human verification.
Apache NiFi 2.0	PostgreSQL 18 SSoT	TCP 5432 / Native	nifi_ingest_writer DB Role	Sole authoritative writer

Source Component	Target Component	Port / Protocol / API Ingress	Security Boundary / Access Key	Operational Significance / Flow Description
Ingest Gate	Store	PostgreSQL		committing Tier 0 Golden SSoT data and bda_provenance metadata tags.
PostgreSQL 18 SSoT Store	Fusio REST / gRPC Gateway	TCP 5432 / Read-Only Views	PostgreSQL Row-Level Security (RLS)	Exposes fine-grained SQL views and endpoints to web applications and BI dashboards.
PostgreSQL 18 SSoT Store	MCP Tool Gateway	TCP 5432 / Vector Search	MCP Tool Scope & Ed25519 Token	Serves vector context and structured database tool capabilities to autonomous AI agents.

2.2 API-Ready: Universal Data Exchange

-  **Bidirectional Data Flow:** Data is no longer trapped inside static visual portals. Enterprise applications, IoT devices, and partner APIs can inject structured telemetry or transactional events directly into our core via REST or gRPC endpoints, as well as extract real-time datasets.
-  **System Interoperability:** Enables zero-friction integration with any third-party, enterprise, or open-source application without requiring vendor-locked connectors.

2.3 MCP-Ready: Native AI & LLM Capability

-  **Direct Agent Integration:** Implements the open standard **Model Context Protocol (MCP)**. Large Language Models (LLMs) and autonomous AI agents can directly query database metrics, trigger background transformations, and retrieve vector embeddings as native “Tools”.
-  **Contextual Grounding:** Replaces static PDF/Excel exports with conversational, context-aware AI interactions connected directly to live database state.

2.4 Fine-Grained Access Control (FGAC) & Data Tagging Governance

-  **Row and Column Level Security:** Permissions are strictly enforced at the API gateway and PostgreSQL database layer using session context injection (SET LOCAL). An external application or AI agent accesses only the precise data slices authorised for its identity.

- 🔗 **Human SSoT & AI Provenance Metadata Tagging:** To ensure total data authenticity and governance, all data within the Big Data Analytics Lakehouse is partitioned into two distinct categories:
 - Real Data & Human Verification (Tier 0 SSoT):** Human-entered data is validated through the Laravel human-in-the-loop portal. Apache NiFi 2.0 acts as the sole authoritative writer to Percona Patroni PostgreSQL 18.
 - AI Processes Enriched with RAG & Generative Metadata:** Any dataset touched, generated, or enriched by AI agents is explicitly tagged using bda_provenance metadata. This metadata records cryptographic signature contracts including signature, key_id, verification_status, verification_timestamp, signature_algorithm (Ed25519), and signature_encoding (HEX_RAW_64_BYTE), binding canonical RFC 8785 byte streams.
- 📋 **Auditability & Zero Trust:** Every API call and MCP tool execution is logged, providing clear lineage and governance for regulatory compliance.

3. Financial & Operational ROI

Area	Legacy Architecture (Tableau-based)	Proposed Architecture (API/MCP on Podman)
Licensing Costs	High recurring per-user and per-core licensing fees.	Zero proprietary licensing fees (100% Open-Source).
Data Accessibility	Locked inside proprietary .twb workbooks and visual dashboards.	Universal API & MCP endpoints accessible by any tool or agent.
Ingestion Capability	Primarily unidirectional read-only reporting.	Fully bidirectional (Ingest, Transform, Export, Query).
AI Integration	None (Manual exports required for AI context).	Native MCP support for autonomous AI workflows.
Security Granularity	Dashboard-level / Workbook-level permissions.	Fine-Grained Access Control (Row/Column/Tool level).

4. Tableau Migration & Modernisation Strategy

To ensure zero downtime and manage operational risk, legacy Tableau workbooks will be systematically decommissioned using a four-phase migration roadmap:

[Phase 1: Audit] → [Phase 2: Logic Transfer] → [Phase 3: Open BI] → [Phase 4: MCP/API]

Phase 1: Workbook Audit & Inventory

- Catalogue all active Tableau workbooks (.twb/.twbx), calculated fields, custom SQL scripts, and user access lists.
- Identify redundant reports and mark high-value dashboards for migration.

Phase 2: Data & Logic Consolidation

- Migrate complex Tableau calculations and data blending logic into **PostgreSQL Materialised Views** and stored functions.
- Ensure Apache NiFi orchestrates data pipelines directly into clean PostgreSQL schemas.

Phase 3: Open-Source BI Deployment

- Deploy containerised **Apache Superset** (or Metabase) on Podman to replicate essential executive dashboards.
- Connect directly to the PostgreSQL layer, restoring visual reporting capabilities with zero user-license overhead.

Phase 4: API & MCP Enablement

- Expose underlying business calculations as REST/gRPC API endpoints.
- Wrap PostgreSQL metrics and vector searches into standardised **MCP Tools** for internal AI agent consumption.

5. Podman Infrastructure & Container Blueprint

The target infrastructure relies on rootless **Podman** pods to enforce high availability, zero vendor lock-in, and full Kubernetes compatibility.

```
# Conceptual Architecture Blueprint: podman-pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: bda-ai-infra-pod
spec:
  containers:
    - name: postgres-engine
      image: docker.io/pgvector/pgvector:pg18
      description: "Central database store equipped with
vector search capabilities."

    - name: nifi-orchestrator
      image: docker.io/apache/nifi:latest
      description: "Low-latency data ingestion, batch
processing, and ETL orchestrator."
```

```
- name: mcp-api-gateway
  image: localhost/bda-mcp-server:v1
  description: "Custom Python/Node.js gateway delivering
Open APIs, MCP Tools, and FGAC enforcement."

- name: open-bi-superset
  image: docker.io/apache/superset:latest
  description: "Open-source business intelligence platform
replacing Tableau."
```

6. Next Steps & Execution Plan

Upon approval of this proposal, execution will proceed as follows via automated code and configuration updates:

1. **Commit Proposal Document:** Save docs/IT-MANAGEMENT-PROPOSAL.md into the main branch.
2. **Deploy Podman Pod Spec:** Add docker/podman-pod.yaml containing the complete container definition.
3. **Build MCP Server Skeleton:** Commit the Python-based MCP server in src/mcp-server/ with initial PostgreSQL tool connections and FGAC middleware.
4. **Initiate Phase 1 Migration:** Begin Tableau workbook audit and SQL logic extraction.